

ELEC9731, 2023, Term 1, ASSIGNMENT 2

due – week 9 Tuesday 11/04/2020 17:00 (Sydney time).

submission – please submit as a single **pdf** file by email
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use subject: ELEC 9731-Assignment 2

Late assignments will be penalised at 10% of the maximum value per day late

Question 1 (8 marks). A linear control system is described by the equations

$$\begin{aligned}\dot{x}(t) &= \begin{pmatrix} 1.8 & 0 & 5a \\ a & 3a^3 & 8.0 \\ -3.0 & 0 & 7.0 \end{pmatrix} x(t) + \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} u(t), \\ y(t) &= \begin{pmatrix} -2a^2 & 0 & 3.7 \\ 4.6 - a & a - 4 & 3.8 \\ 0.6 & 0 & 0.9 \end{pmatrix} x(t),\end{aligned}$$

where a varies from $-\infty$ to ∞ . Determine the range of a for observability.

Question 2 (9 marks). Consider the following systems of two linear second order ODEs:

$$\begin{aligned}\ddot{z}_1(t) &= -z_1(t) + az_2(t) + u_1(t), \\ \ddot{z}_2(t) &= -z_2(t) + az_1(t) - 2u_2(t),\end{aligned}$$

where a is some parameter.

- (i) Re-write this system in a state space form.
- (ii) For what values of a this system is controllable?
- (iii) For what values of a this system is controllable from u_1 alone?
- (iv) For what values of a this system is controllable from u_2 alone?
- (v) If the controls are constrained to be equal, $u_1 = u_2$, for what values of a this system is controllable?
- (vi) Find the transfer function from u to $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$. Hence find the the Smith-McMillan form.

Question 3 (8 marks). The characteristic polynomial of a control system is the following uncertain polynomial:

$$s^3 + a_2 s^2 + a_1 s + 3$$

where $a_1 \in [1, 4]$, $a_2 \in [1, 4]$ and $5 - K \leq a_1 + a_2 \leq 5 + K$ where $K \in [0, +\infty)$.

For what values of K is this uncertain polynomial stable?